

## 2.4 Given the following signals

$$x(t) = 2\delta(t), \quad y(t) = 4u(t), \quad z(t) = e^{-2t}u(t),$$

Evaluate the following operations.

(a)  $x(t) * y(t)$

(b)  $x(t) * z(t)$

(c)  $y(t) * z(t)$

(d)  $y(t) * [y(t) + z(t)]$

$$\begin{aligned} \text{(a)} \quad x(t) * y(t) &= \int_{-\infty}^{\infty} 2\delta(\tau) \cdot 4u(t-\tau) d\tau \\ &= 8u(t) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x(t) * z(t) &= \int_{-\infty}^{\infty} 2\delta(\tau) \cdot e^{-2(t-\tau)}u(t-\tau) d\tau \\ &= 2e^{-2t}u(t) \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad y(t) * z(t) &= \int_{-\infty}^{\infty} 4u(\tau) \cdot e^{-2(t-\tau)}u(t-\tau) d\tau \\ &= 4e^{-2t} \int_0^t e^{2\tau} d\tau \\ &= \frac{4e^{2\tau}}{2} e^{-2t} \Big|_0^t \\ &= 2(1 - e^{-2t})u(t) \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad y(t) * [y(t) + z(t)] &= y(t) * y(t) + y(t) * z(t) \\ &= \int_{-\infty}^{\infty} 4u(\tau) \cdot 4u(t-\tau) d\tau + \int_{-\infty}^{\infty} 4u(\tau) \cdot e^{-2(t-\tau)}u(t-\tau) d\tau \\ &= 16 \int_0^t 1 d\tau + 2(1 - e^{-2t})u(t) \\ &= 16t + 2(1 - e^{-2t})u(t) \end{aligned}$$

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**2.14** The impulse response of a low-pass filter is  $h(t) = e^{-t}u(t)$ . Determine its step response, that is, the output when the input is a unit step.

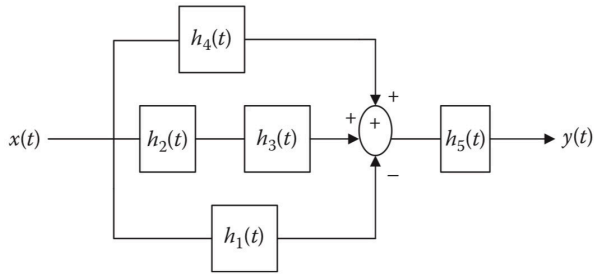
$$\int_{-\infty}^{\infty} e^{-\tau} u(\tau) \cdot u(t-\tau) d\tau$$

$$= \int_0^t e^{-\tau} d\tau$$

$$= -e^{-\tau} \Big|_0^t$$

$$= (1 - e^{-t})u(t)$$

**2.24** Determine the overall impulse response for the system shown in Figure 2.34.



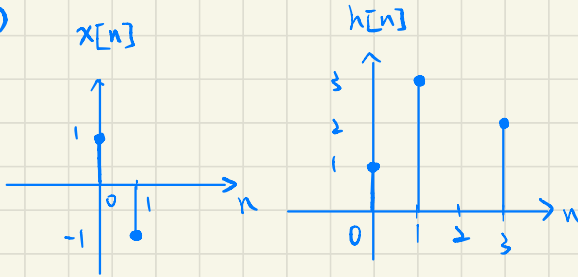
$$h(t) = [h_4(t) + [h_2(t) * h_3(t)] - h_1(t)] * h_5(t)$$

2.30 Given that  $x[n] = \begin{cases} 1, & n=0 \\ -1, & n=1 \\ 0, & \text{otherwise} \end{cases}$ ,  $h[n] = \begin{cases} 1, & n=0 \\ 3, & n=1 \\ 2, & n=3 \\ 0, & \text{otherwise} \end{cases}$

(a) Sketch  $x[n]$  and  $h[n]$ .

(b) Find  $x[n]*h[n]$ .

(a)



(b)

$$x[n] = \delta[n] - \delta[n-1]$$

$$h[n] = \delta[n] + 3\delta[n-1] + 2\delta[n-3]$$

$$x[n] * h[n]$$

$$= x[n] * [\delta[n] + 3\delta[n-1] + 2\delta[n-3]]$$

$$= \delta[n] - \delta[n-1] + 3\delta[n-1] - 3\delta[n-2] + 2\delta[n-3] - 2\delta[n-4]$$

$$= \delta[n] + 2\delta[n-1] - 3\delta[n-2] + 2\delta[n-3] - 2\delta[n-4] *$$

### 2.33 Two systems are described by

$$h_1[n] = (0.4)^n u[n], \quad h_2[n] = \delta[n] + 0.5\delta[n-1]$$

Determine the response to the input  $x[n] = (0.4)^n u[n]$  if

(a) The two systems are connected in parallel

(b) The two systems are connected in cascade

$$(a) \quad x[n] * [h_1[n] + h_2[n]]$$

$$= x[n] * h_1[n] + x[n] * h_2[n]$$

$$= \sum_{k=-\infty}^{\infty} (0.4)^k u[k] \cdot (0.4)^{n-k} u[n-k] + \sum_{k=-\infty}^{\infty} (0.4)^k u[k] (\delta[n-k] + 0.5\delta[n-1-k])$$

$$= (0.4)^n \sum_{k=0}^n 1 + (0.4)^n u[n] + 0.5(0.4)^{n-1} u[n-1]$$

$$= (n+1)(0.4)^n u[n] + (0.4)^n u[n] + 0.5(0.4)^{n-1} u[n-1]$$

$$= (n+2)(0.4)^n u[n] + 0.5(0.4)^{n-1} u[n-1] \neq$$

$$(b) \quad x[n] * [h_1[n] * h_2[n]]$$

$$= x[n] * [(0.4)^n u[n] + 0.5(0.4)^{n-1} u[n-1]]$$

$$= (0.4)^n u[n] * [(0.4)^n u[n] + 0.5(0.4)^{n-1} u[n-1]]$$

$$= \sum_{k=-\infty}^{\infty} (0.4)^k u[k] \cdot (0.4)^{n-k} u[n-k] + \sum_{k=-\infty}^{\infty} (0.4)^k u[k] \cdot 0.5(0.4)^{n-1-k} u[n-1-k]$$

$$= (n+1)(0.4)^n u[n] + 0.5(0.4)^{n-1} \sum_{k=0}^{n-1} 1$$

$$= (n+1)(0.4)^n u[n] + 0.5n(0.4)^{n-1} u[n-1] \neq$$

**2.36** The input  $x[n] = [1 \ -1]$  to a system produces the output  $y[n] = [4 \ 2 \ 5 \ 1]$ . Determine the impulse response.

$$y[n] = h[n]x[0] + \sum_{k=0}^{n-1} h[k]x[n-k]$$

$$h[n] = \frac{1}{x[0]} \left[ y[n] - \sum_{k=0}^{n-1} h[k]x[n-k] \right]$$

$$h[0] = \frac{y[0]}{x[0]} = 4$$

$$h[1] = 2 - 4 \times (-1) = 6$$

$$h[2] = 5 - 4 \times 0 - 6 \times (-1) = 11$$

$$h[3] = 1 - 11 \times (-1) = 12$$

$$h[4] = 0 - 12 \times (-1) = 12$$

$\vdots$

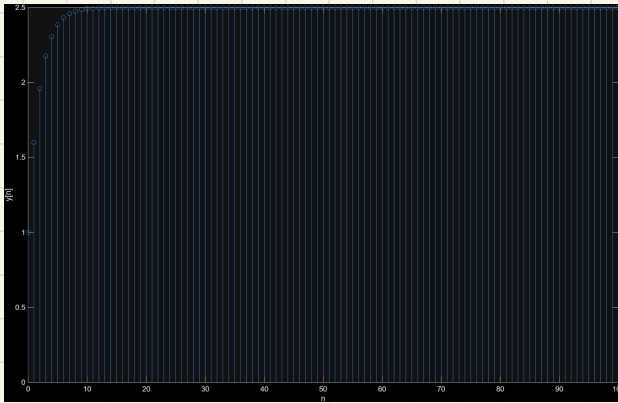
$$h[n] = [4, 6, 11]$$

**2.39** An LTI discrete system has the impulse response  $h[n] = (0.6)^n u[n]$ . Use MATLAB to calculate the response of the system to input  $x[n] = u[n]$  and plot it.

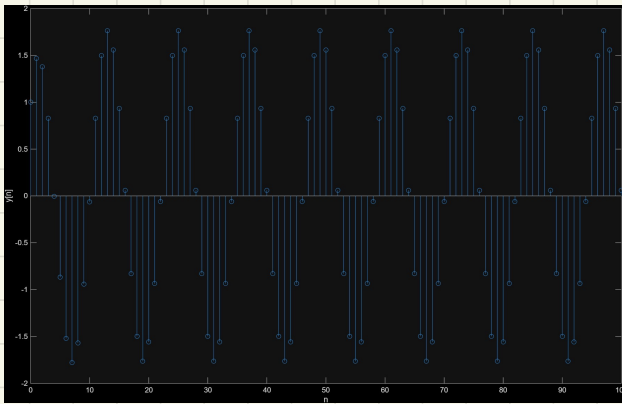
**2.40** Repeat the previous problem for  $x[n] = \cos(n\pi/6)u[n]$ .

**2.41** Given that  $x[n] = [1 \ -1 \ 2 \ 4]$  and  $y[n] = [2 \ 6 \ 4 \ 0 \ 8 \ 5 \ 12]$ , use MATLAB to find  $h[n]$ .

2.39



2.40



2.41  
(a)

$$h[0] = \frac{y[0]}{x[0]} = 2$$

$$h[1] = \frac{1}{x[0]} [y[1] - x[1]h[0]] = 6 - (-1) \times 2 = 8$$

$$h[2] = 4 - (-1) \times 8 - 2 \times 2 = 8$$

$$h[3] = 0 - (-1) \times 8 - 2 \times 8 - 4 \times 2 = -16$$

$$h[n] = [2, 8, 8, -16]$$

(b) use deconv() in Matlab

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>> HW2_2_41
Impulse response h[n] is :
    2     8     8    -16
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